La Scuola d'Italia / Dr. Huson / IB Math 11
27 October 2026 Markscheme

2.5 Pre-Quiz: Sequences, Lines, Exponents and Radicals — Markscheme

Total: 58 marks.

1. [Maximum mark: 10]

(a) $u_1 + 3d = 23, \quad u_1 + 10d = 51$ **M1**

$7d = 28, \quad d = 4$ **A1**

$u_1 = 23 - 3(4) = 11$ **A1**

M1 for two equations from $u_n = u_1 + (n-1)d$, or for $d = \frac{51-23}{11-4}$. Wrong method: $u_1 = 23 - 4(4) = 7$ uses n in place of $n-1$: A0 for u_1.

(b) $u_n = 11 + 4(n-1)$ **M1A1**

Accept $u_n = 4n + 7$. FT from (a).

(c) $S_{20} = \frac{20}{2}\left(2(11) + 4(20-1)\right)$ **M1**

$S_{20} = 980$ **A1**

Accept $\frac{20}{2}(11 + 87)$. Wrong method: $u_{20} = 87$ is the twentieth term, not the sum: M0A0. FT from (a).

(d) $11 + 4(n-1) = 199$ **M1**

$4(n-1) = 188, \quad n = 48$ **A1**

n is a whole number, so 199 is a term: the 48th term **R1**

R1 for stating that n is a whole number (a positive integer). Wrong method: $11 + 4n = 199$ gives $n = 47$: M0A0, R1 may still be awarded. FT from (b).

2. [Maximum mark: 8]

(a) $u_1 r = 12, \quad u_1 r^4 = 324, \quad \text{so } r^3 = \dfrac{324}{12} = 27$ **M1**

$r = 3$ **A1**

$u_1 = \dfrac{12}{3} = 4$ **A1**

M1 for dividing the two terms. Wrong method: $r = \frac{324}{12} = 27$ *without the cube root: M1A0A0.*

(b) $S_8 = \dfrac{4\,(3^8 - 1)}{3 - 1}$ **M1**

$S_8 = 13\,120$ **A1**

Wrong method: $u_8 = 8748$ *is the eighth term: M0A0. FT from (a).*

(c) $\dfrac{4\,(3^n - 1)}{3 - 1} > 100\,000$ **M1**

$S_9 = 39\,364, \quad S_{10} = 118\,096$ **A1**

$n = 10$ **A1**

M1 for an inequality or a GDC table of S_n*. A1 for the values either side of* $100\,000$*, or for the graph intersection* $n = 9.85\ldots$ *seen. A final answer of* 9.85 *earns A0. Wrong method: using* u_n *in place of* S_n *gives* $n = 11$*: M0A0A0. FT from (a).*

3. [Maximum mark: 11]

(a) $m_{PQ} = \dfrac{3-8}{6-(-4)}$ **M1**

$m_{PQ} = -\dfrac{5}{10} = -\dfrac{1}{2}$ **A1**

A bare answer earns M0A1.

(b) $m_L = 2$ **A1**

$y - 9 = 2(x-4)$ **M1**

$y = 2x + 1$ **A1**

A1 for the negative reciprocal of their gradient. The final A1 needs the form $y = mx + c$. Wrong method: $m_L = -2$ (reciprocal without the sign change) gives $y = -2x + 17$: A0M1A0. FT from (a).

(c) line PQ: $y - 3 = -\frac{1}{2}(x-6)$, so $y = -\frac{1}{2}x + 6$ **A1**

$2x + 1 = -\frac{1}{2}x + 6$ **M1**

$x = 2, \quad y = 5 \qquad F(2,\ 5)$ **A1A1**

M1 for solving their two lines together by any method (equating, substitution, GDC). Wrong method: the midpoint of PQ, $(1,\ 5.5)$, is not the foot of the perpendicular from R: A1 for the equation of PQ if seen, then M0A0A0. FT from (b).

(d) $RF = \sqrt{(4-2)^2 + (9-5)^2} = \sqrt{20}$ **M1**

$RF = 2\sqrt{5}$ **A1**

Accept $a = 2$. $\sqrt{20}$ or 4.47 alone earns A0. Wrong method: $RP = \sqrt{65}$ or $RQ = 2\sqrt{10}$ is not the perpendicular distance: M0A0. FT from (c).

4. [Maximum mark: 7]

(a) $3x + 2y = 61$ **A1**

$2x + 5y = 70$ **A1**

(b) $6x + 4y = 122$ and $6x + 15y = 210$, so $11y = 88$ **(M1)**

adult ticket $x = 15$ euros **A1**

child ticket $y = 8$ euros **A1**

M1 for any valid method, including the GDC simultaneous-equation solver; a correct GDC answer with no other working earns full marks. FT from (a).

(c) $4(15) + 8k = 156$ **M1**

$8k = 96, \quad k = 12$ children **A1**

Wrong method: swapping the prices, $4(8) + 15k = 156$, *gives* $k = 8.27$, *not a whole number: M0A0. FT from (b).*

5. [Maximum mark: 12]

(a) $(3a^2b^{-1})^2 = 9a^4b^{-2}$ **M1**

$9a^4b^{-2} \times 2a^{-3}b^4 = 18a^{4-3}b^{-2+4}$ **A1**

$= 18ab^2$ **A1**

M1 for raising every factor of the bracket to the power 2. *Wrong method: leaving the* 3 *unsquared gives* $6ab^2$*: M0A0A0.*

(b) $\left(\frac{16}{81}\right)^{-\frac{3}{4}} = \left(\frac{81}{16}\right)^{\frac{3}{4}} = \left(\frac{3}{2}\right)^3$ **M1**

$= \frac{27}{8}$ **A1**

Accept $3\frac{3}{8}$. 3.375 *from the GDC with no working earns M0A1. Wrong method: ignoring the negative sign gives* $\frac{8}{27}$*: M0A0.*

(c) $(2^2)^{2x-1} = (2^3)^{x+2}$ **M1**

$4x - 2 = 3x + 6$ **A1**

$x = 8$ **A1**

M1 for writing both sides as powers of 2. *Wrong method: equating the exponents without changing the base,* $2x - 1 = x + 2$*, gives* $x = 3$*: M0A0A0.*

(d) $u^2 - 10u + 9 = 0, \quad (u-1)(u-9) = 0$ **M1**

$u = 1$ or $u = 9$ **A1**

$3^x = 1 \Rightarrow x = 0$ **A1**

$3^x = 9 \Rightarrow x = 2$ **A1**

M1 for the quadratic in u *(using* $3^{2x} = u^2$*) solved by any method. Stopping at* $u = 1, 9$ *earns M1A1A0A0. Wrong method: reading* 3^x *as* $3x$ *gives* $x = \frac{1}{3}, 3$*: A0A0 for those values.*

6. [Maximum mark: 10]

(a) $7\sqrt{2} - 4\sqrt{2} + 3\sqrt{2}$ **M1**

$= 6\sqrt{2}$ **A1**

M1 for writing at least two of the surds as multiples of $\sqrt{2}$*. Wrong method:* $\sqrt{98 - 32 + 18} = \sqrt{84}$*: M0A0.*

(b) $\dfrac{10}{\sqrt{7} - \sqrt{2}} \times \dfrac{\sqrt{7} + \sqrt{2}}{\sqrt{7} + \sqrt{2}}$ **M1**

$= \dfrac{10\left(\sqrt{7} + \sqrt{2}\right)}{7 - 2}$ **A1**

$= 2\sqrt{7} + 2\sqrt{2}$ **A1**

A1 for the denominator 5*. Accept* $a = 2$*,* $b = 2$*. Wrong method: denominator* $7 + 2 = 9$*: M1A0A0.*

(c) $x\sqrt{3} - x = 2, \quad x\left(\sqrt{3} - 1\right) = 2$ **M1**

$x = \dfrac{2}{\sqrt{3} - 1} \times \dfrac{\sqrt{3} + 1}{\sqrt{3} + 1} = \dfrac{2\left(\sqrt{3} + 1\right)}{2}$ **A1**

$x = 1 + \sqrt{3}$ **A1**

M1 for collecting the x*-terms and factorizing. A1 for multiplying by the conjugate. Accept* $a = 1$*,* $b = 1$*.* 2.73 *earns A0A0. Wrong method:* $x\sqrt{3} - x = x\sqrt{2}$*: M0A0A0.*

(d) $\left(4 + \sqrt{5}\right)\left(4 - \sqrt{5}\right) = 16 - 5$ **M1**

Area $= 11\,\text{cm}^2$ **A1**

Wrong method: $16 + 5 = 21$ *(sign of the product of the surds): M1A0.*